

## **Integrated Teaching Model: A Follow-Up with Fundamental Aerodynamics**

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# Integrated Teaching Model: A Follow Up with Fundamental Aerodynamics

Sidaard Gunasekaran<sup>1</sup>

## Abstract

The integrated teaching model was implemented for the second time in the undergraduate fundamental aerodynamics class at the University of Dayton. As with any aerospace classes, the fundamental aerodynamics relies heavily on mathematics. The integrated teaching model is specially designed to let the students think beyond the equations and understand *through experience* the applications and impact of the concepts and equations in real life. Both student-centered and teacher-centered experiential learning was heavily incorporated in each aspect of the class which made a monumental difference in the students' understanding of the subject. The scaffolding of the class structure and the open-ended homework assignments helped students to acquire multiple technical skillsets in experimental and computational aerodynamics. Some of the skillsets include designing airfoil using Joukowski transformation, performing inviscid flow simulation using panel methods on the airfoil, 3D printing the airfoil and testing it in the wind tunnel, numerically determining vorticity and circulation of wingtip vortex, leading edge vortex, etc. Through the passion project component of the class, students were able to perform experiments on plethora of applications of aerodynamics such as golf drivers, speed chutes, sailboats, golf ball, propellers, drones, delta wings, airplanes, etc. ([goo.gl/1vJwGF](https://goo.gl/1vJwGF)). Coupled with an assessment technique where the students are required to integrate homework, projects, activities, lectures and independent studies on a single platform (portfolio), the model promotes sustainable learning (long-term learning), communication skills, technical skills and critical thinking skills in a way that a conventional lecture-based model wouldn't. Multiple Likert scale assessments of the modules and qualitative feedback from the students will be shared in this paper along with homework assignments and projects.

## 1. Introduction

At the University of Dayton, the subject of Aerodynamics is taught as a required senior-level undergraduate course to expose students of aerospace concentration or minor to governing equation of aerodynamics, especially focused on the inviscid theory. The course is usually offered in the spring semester of each academic year. This leads to coverage of topics such as Laplace equations, derivation, and combination of elementary flows, potential flow theory, thin airfoil theory, lifting line theory, vortex panel methods, etc. As evident from these topics, the subject of aerodynamics (at least in this context) is theory-rich and experience poor. The prerequisite for this class is a junior level fluid dynamics course where the students are already exposed to the governing equations of fluid dynamics. Students usually take Aerodynamics course one semester or two semesters after they take fluid dynamics. But in author's experience, on an average, students are scared of the governing equation and regard them as complicated equations which they are uncomfortable with. It is mainly because the fluid dynamics is heavily based on the theory and the connection between the equations and the real life was not fostered. Therefore, if the application side of aerodynamics is masked, the course will resemble a basic mathematics course. John Dewey [1] expressed his belief that subject matter should not be learned in isolation, and that education should begin with student experience and should be contextual. As instructors, there is always a struggle to cultivate students experience and use it to provide context for the concepts, especially in a theory-rich, math-heavy classes such as Aerodynamics. In each class, students and the instructor writes several pages

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of equations deriving the above-mentioned theories. With the plethora of modern active learning techniques available, the instructor faces another challenge to select an appropriate technique which can be used in this type of class not only to keep the students engaged but also to convey the significance of the equations and make relevant connections to foster understanding. Even when active learning strategies are employed, the role of teachers in this type of classroom usually follows the “banking model” of education where they deposit knowledge into the empty repository of the student mind (Freire [2], Gattegno [3]).

Even though the fundamental theories are of great importance, without an overall understanding and application of the equations and the concepts, the subject of aerodynamics can easily be deemed as “boring” by the students even though it is one of the most exciting subjects in the field of aerospace engineering and has its roots set all the way back in time to Aristotle in 350 B.C. [4]. Aerodynamics has variety of applications, beyond airplanes. Any application in a conventional aerodynamics class comes from students solving back of the chapter homework problems. Students use the equations they learned in class to a hypothetical problem to get an answer by inserting numbers into the equation. One such question from Fundamentals of Aerodynamics book by John Anderson [5] (which is adopted by most instructors as a required textbook for aerodynamics course) is given below:

*Consider a NACA 2412 airfoil. The airfoil is flying at a velocity of 60 m/s at a standard altitude of 3 km. The chord length of the airfoil is 2m. Calculate the lift per unit span when the angle of attack is 4°.*

In this problem, students have to employ the equation of lift per unit span  $L'$

$$L' = \frac{1}{2} \rho V^2 c C_l \quad (1)$$

where  $\rho$  is density,  $V$  is velocity,  $c$  is chord length and  $C_l$  is the sectional coefficient of lift. They have to look up the value of density at 3 km altitude and the coefficient of lift of NACA 2412 from 4° angle of attack, plug it into equation 1 and determine lift per unit span. Even though this type of problems allows the students to practice the equations, it doesn't necessarily invoke critical thinking or understanding of any sort. Even though there are questions in the textbook which presents the problem from a context of an airplane, the book rarely applies the concepts to other applications of aerodynamics.

Therefore, it was clear to the author that if this subject had to make an impact in students understanding and interest in the field of aeronautical engineering, it cannot be done through conventional lecture-based model. On the other hand, the fundamental theories cannot be overlooked in the interest of focusing on applications. This paper documents an endeavor to achieve a balance between those two by following McLaren's [6] suggestion that the theory informs practice but experiential and practical knowledge can be employed as a means to understanding and interpreting that theory. The objective is to approach a math-based Aerodynamics class from a context of application, student-centered (SCEL) and instructor-centered experiential learning (ICEL) at the same time having roots of those EL in the theoretical formulations. The changes that were made in this course was certainly noticed by the students as one puts it:

*“My ability to appreciate the math going on in aerodynamics is largely in part due to our learning style in class. After talking with a student who previously took aerodynamics, I have found a new appreciation for how we went through the semester. The student I talked to said his professor just went through proofs quickly and the student was very confused, not really understanding the concepts behind the topics they discussed. But in Sid's class, not only have I been given a strong foundation for aerodynamics, but the handouts and resources we have been provided are ones that I can use in the future if I ever need to review the fundamentals of aerodynamics, and I am very grateful that I have them.”*

*- Madeline L*

Apart from using multiple active learning techniques some of which are detailed in the paper, the integrated teaching pedagogy detailed in Gunasekaran [7] was employed. A brief overview of the pedagogy is described below.

## 2. Integrated Teaching Model

The motivation behind developing the integrated teaching pedagogy was to incorporate the necessary skillsets proposed by National Institute of Literacy (NIL) for a common man to be successful in the 21<sup>st</sup> century in a math-based engineering class. The objective of teaching model is to incorporate five major skillsets which were inspired from EEF standards [8]:

1. Applying subject concepts in real world
2. Solving open-ended problems
3. Performing critical thinking
4. Engaging in independent research
5. Practicing technical communication

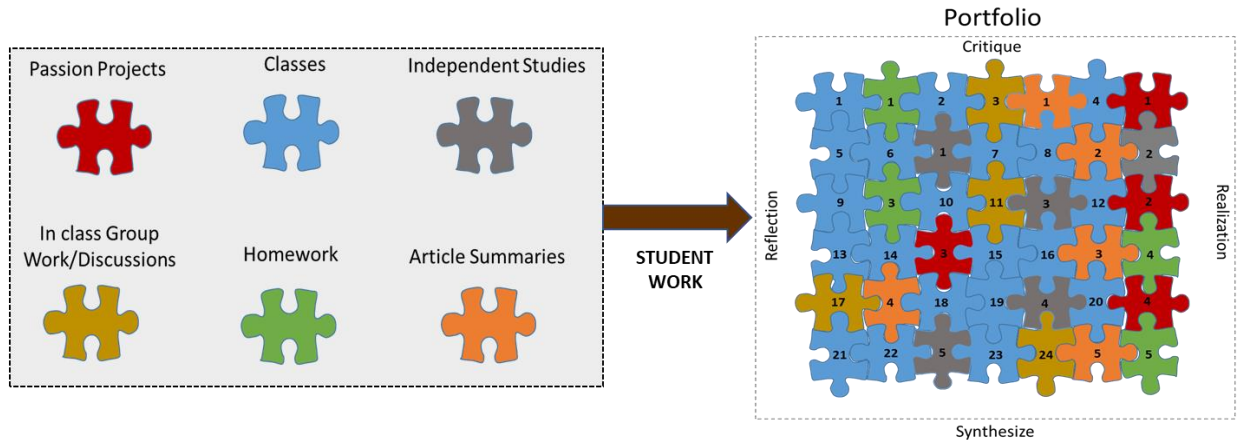
In the integrated teaching model, portfolio is used as a platform to integrate and connect multiple course elements such as lectures, assignments, projects, article summaries, independent studies, and students' reflections throughout the course of the semester. The intention is to make the relevant and required connections between the concepts and topics which foster better understanding of the subject when compared to the monotonous progression of well-structured chapters in the textbook. In the portfolio, students are required to employ the Feynman technique where they explain fundamental concepts using simple words. They are also required to make connections between the different aspects of the classes. Through the process of integration of these multiple entities of a course, students learn to critique, realize, synthesize and reflect on the subject they learn thereby achieving all the stages of Bloom's taxonomy.

*"Reflecting on this semester, there are many things I have learned and will stick with me because of the way this class was arranged. I believe passion projects and portfolios were beneficial to my understanding of the subject and the questions asked in homework were challenging but fair, a combination that most professors should shoot for if they want to increase students' understanding of anything. I also think it was beneficial having a professor who was passionate about the subject he was teaching and took a genuine interest in our learning."*

- Jared M

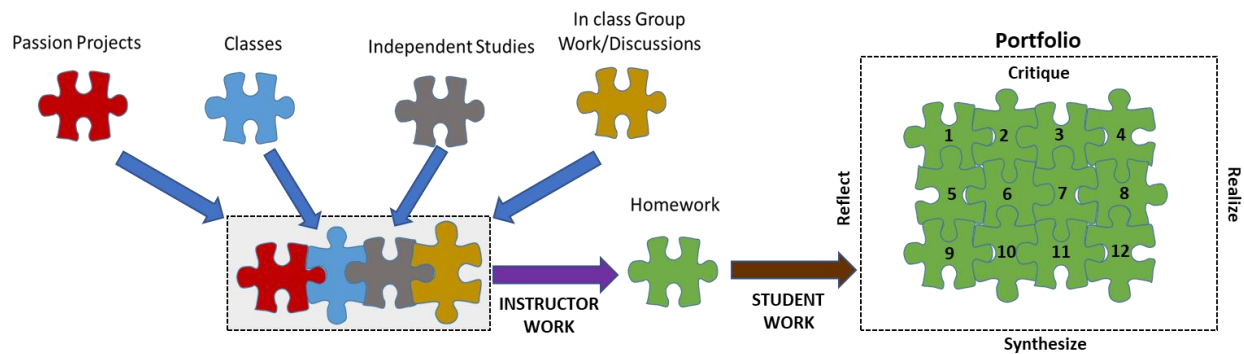
## 3. Modified Integrated Teaching Model

The integrated teaching model was first implemented in a graduate compressible aerodynamics class at the University of Dayton in the Fall of 2016. The challenges of this teaching model was documented in Gunasekaran [7]. One of the major challenge in the first trial was the amount of workload for the students and the instructor. A schematic of the integrated teaching pedagogy is depicted in Figure 1. As the schematic indicates, the role of instructor in the model is to teach the classes, assign projects and homework independently. As mentioned before, in each portfolio iteration (usually once every two weeks), students had to explain their understanding of concepts from four classes, explain the work they did in one homework assignment which usually involves solving open-ended problem and perform independent study on a minimum of two topics. They were also required to find connections between the content of the class, homework and the independent study in such a way that the portfolio has a logical progression of ideas and student thought process and not sporadic categorization of homework and classes. In fact, the students were instructed NOT to organize the portfolio in terms of classes, homework and independent study. The general narrative of the portfolio content should contain all elements shown in Figure 1.



**Figure 1 Schematic of the Integrated Teaching Pedagogy.**

In the end of Spring 2017 semester, each student in the graduate compressible flow class had a portfolio of more than 100 pages (on an average) containing a detailed description of the concepts in the subject. They summarized content in 23 classes, 5 homework assignments, 5 independent research, and 5 article depicted as mentioned in Figure 1. Due to the nature of the subject, experiential learning was not employed since it will require a supersonic wind tunnel. But additional elements such as passion projects and in-class group discussions can be incorporated in the portfolio. Apart from summary of these elements from class, the portfolio also contained detailed reflections from the students. Even though the portfolio in this format brings forth high levels of learning, the amount of workload from the students to complete this task was greater than the time students are required to spend on a three credit undergraduate course per the university standard.

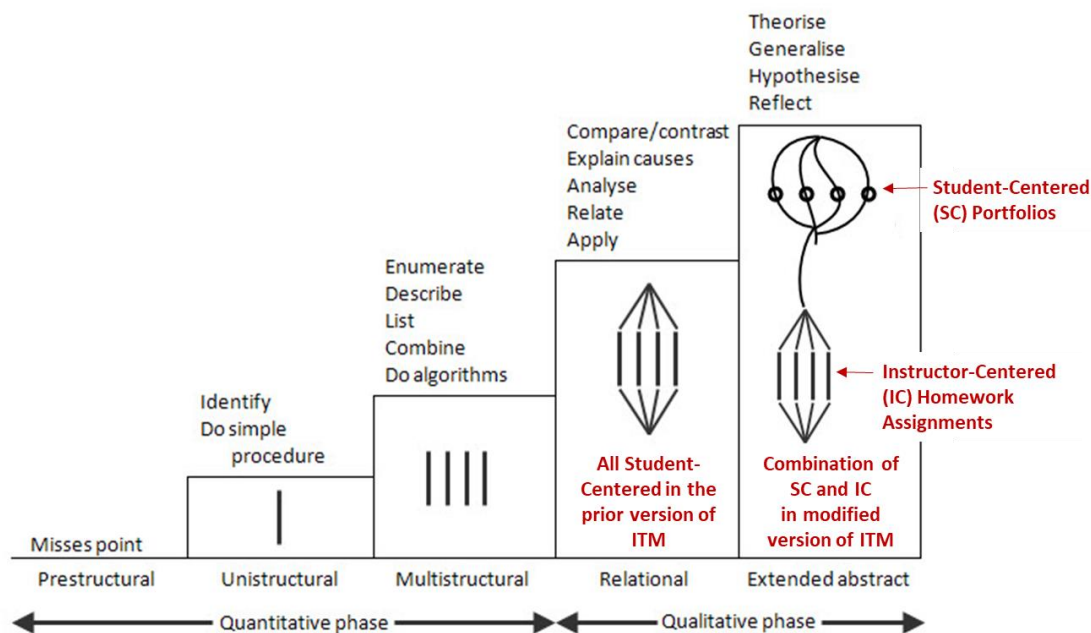


**Figure 2 Schematic of the updated version of the integrated teaching pedagogy.**

This drawback was overcome in the updated model practiced in the aerodynamics course by incorporating questions related to class discussion, open-ended problem, independent study and project into a homework assignment what was assigned each week. The 12 homework assignments that were assigned to the students is shown in the Appendix of this paper for easy access to the readers. The compilation of each homework was done by the instructor each week, therefore the amount of work load the instructor experienced was decreased. The modified schematic of the integrated teaching pedagogy is shown in Figure 2. In this model, the students were required to find connections between each question posed in the homework and to find connections from one homework to the other. Similar to the prior version of the model, the students were specifically instructed NOT to categorize the content of the portfolio in terms of homework numbers (Homework 1, Homework 2, etc.,). Their general narrative in the portfolio should contain answers to the questions posed in each homework assignments which should have logical

progression of ideas. The modified teaching model decreased the overall time that was spent by each student towards creating the portfolio and didn't have major impact on the learning and understanding of the subject by the students.

One of the main differences between the original and the modified version of the integration teaching model is that the original teaching model was entirely student-centered whereas the modified version of the model is both instructor-centered and student-centered where the initial scaffolding is done by the instructor and the students put the “puzzles” together themselves. The modified teaching model shows some resemblance to one of the popular teaching model proposed by Collis and Higgs [9] called the Structure of Observed Learning Outcomes (SOLO) taxonomy. The SOLO taxonomy was through careful analysis of multiple student responses to assessment tasks and has been validated for use in wide range of disciplines. There are five stages in the SOLO taxonomy which are shown in Figure 3.



**Figure 3 shows the five stages of SOLO taxonomy [9] dictating the process to achieve deep learning from surface learning. The relational stage of the taxonomy is similar to the prior version of the integrated teaching model (ITM) [7] whereas the modified ITM consists of combination of student-centered and instructed-centered learning.**

The first level of SOLO is a stage of ignorance that exists outside of the taxonomy. The next two stages (unistructural and multi-structural) are both levels of surface understanding in which knowledge is accumulated in greater quantity. Simple accumulation of knowledge doesn't result in an in-depth understanding of the subject. The in-depth understanding comes with a qualitative change in how the ideas are understood in connection with other ideas. These connections require increased abstraction from the students. Therefore, the last two stages are characterized by the integration and connection of knowledge as well as abstraction.

The process of getting from multistructural stage to the relational stage of the SOLO taxonomy can happen in two ways:

1. Instructor-centered where the course instructor provides materials necessary for the students to make the connections.
2. Student-centered where the students are required to fit the pieces of the puzzle themselves to make the connections.

In the previous version of the integrated teaching model, the instructor required the students to make the connections among multiple elements of the course in the portfolio. However, in the modified version of the integrated teaching model, the instructor makes the connection by scaffolding multiple elements of the course such as ideas discussed in class, provoking questions requiring independent study, mathematical analysis of an open-ended problem, programming etc., where a common link to a certain concept can be found. By performing the homework assignment, students realize the connection within the homework set by the instructor but also find connections within the homework assignments which were not set by the instructor.

*“Overall, this class was awesome. As my first real class in aerospace engineering it has only brighten and futured my interest in Aerospace applications. I really liked the way the homework maps exactly to what we are doing in class because it makes you learn the material we are learning in class the right way. I have always considered myself to be a learner by doing and finally having a class in college where I am allowed to implement the things I learn in the class room is very exciting and exactly what I looked forward to getting out of this class at the beginning of the semester.”* - Chris B

Through this modified teaching model, the students’ portfolio consisted of pockets of realizations and reflections made by the students along with the connections enforced by the instructor. But doing this required the students to do a lot more work than a conventional course similar to the previous model. But the modified model had more profound impact when compared to the previous version of the model as one student puts it:

*“There are times when I hated this class, namely whenever I spent copious amounts of time on it, not, mind you, because I did not want to learn the material but rather, because I did not have enough time. This class took most of my week. Sometimes I felt like I was in the graduate section. Yet, that is not necessarily a bad thing. One of my main complaints since coming to college is that I never feel like I am being pushed enough. That doesn’t mean that I want to go around frustrated and confused because professors are making concepts harder than they need to be. Rather I want professors to push me individually to become better. I get that from Aerodynamics.”* - Heather L

The passion projects were another aspect of the class where the combination of instructor-centered and student-centered learning proved to be effective. In the passion projects, students can test any application of aerodynamics, study them, perform wind tunnel testing and data analysis using the tools and equations learned in the class. Even though some of the aerodynamic concepts are presented from an instructor-centered standpoint, students have the opportunity to use what they learned and apply their knowledge through hands-on work.

*“Another highlight for this class for me were the passion projects. I had a lot of fun working in the wind tunnel being able to test and apply what we learned in class. The second project we did, creating Joukowski Airfoils was my favorite of the two. Just learning about the Joukowski transformation was fascinating, and I enjoyed being able to apply it even more. I had a role in testing and picking which transformation we would use for three of the airfoils. Although we had many struggles with the wind tunnel gremlins I enjoyed my time experimenting. It was a very nice way to wrap up the semester.”* - Jen C

More details on the passion project aspect of the course is presented in the later sections of the paper.

### 3. Student Learning Outcomes

The teaching pedagogy is said to be effective only when the learning outcomes of the course are met. Employing a certain teaching pedagogy without specific learning outcomes is analogous to taking a journey without knowing the destination. Employing modern teaching pedagogies can result increase in students' engagement and learning but doesn't necessarily guarantee students are learning what they are supposed to learn in a given course.

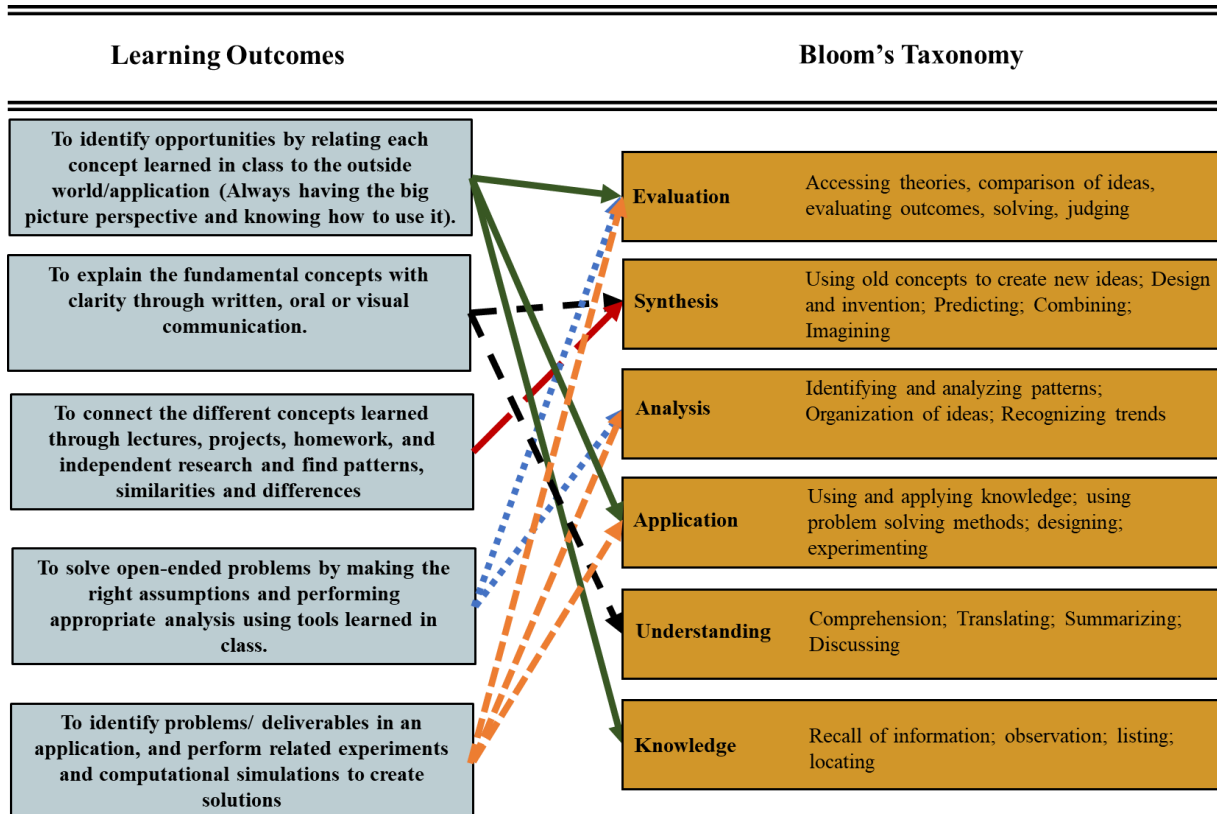


Figure 4 Comparison of Learning Outcomes of Fundamental Aerodynamics class with Bloom's taxonomy

Five major student learning outcomes for the aerodynamics course conceived by the author is shown in Figure 4. The student learning outcomes are designed to be broader so that it can be applied to most of the math based aerospace classes the author teaches. In Figure 4, the learning outcomes are matched with the different levels of Bloom's taxonomy [9]. As seen in the figure, the learning outcomes satisfies all the levels of Bloom's taxonomy. For any learning outcomes to be effective, it needs to be observable and measurable. Even though these learning outcomes are broad, the portfolio proves to be an excellent platform to assess the students on these five learning outcomes. And most of the homework assignments were designed to achieve these outcomes.

*"I do well in school because I fall in love with the material. As I do my homework, ideas fit together and unlock a deeper understanding of the world. I think that the fundamentals of aerodynamics course caters to this idea, with passion projects and global reflections."*  
- Sarah H

A Likert scale assessment shown in Table 1 was taken in the end of the semester in the aerodynamics course to determine if the set learning outcomes were met. The results from the assessment

indicated that the set learning outcomes were met using the integrated teaching pedagogies and multiple active learning modules in the course. Some of the modules that was used in the aerodynamics course is shown in later sections of the paper.

**Table 1 Likert Scale Assessment on Learning Outcomes**

| Questions                              | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|----------------------------------------|-------------------|----------|---------|-------|----------------|
| I feel I achieved Learning Objective 1 |                   |          |         | 7     | 18             |
| I feel I achieved Learning Objective 2 |                   |          | 1       | 14    | 10             |
| I feel I achieved Learning Objective 3 |                   |          | 1       | 6     | 18             |
| I feel I achieved Learning Objective 4 |                   |          | 4       | 9     | 12             |
| I feel I achieved Learning Objective 5 |                   |          |         | 7     | 18             |

Each learning objective was also matched with Kern Entrepreneurial Engineering Network (KEEN's) proposed skillset/mindset. Being a part of the KEEN community, the department of mechanical and aerospace engineering at the University of Dayton is striving to instill entrepreneurial mindset in students in variety of engineering classes. The learning outcomes set by the author satisfies several adjectives listed as a part of the KEEN's proposed engineering skillset [9]. These five learning objectives are used by the author in all the aerospace related classes such as compressible flow aerodynamics, turbulence, introduction to flight, etc.

## 4. Technical Outcomes

One advantage of having broader learning outcomes is that it can be used for multiple courses. The technical competency of the students in a course such as aerodynamics is extremely important. Therefore, the students need to be trained to acquire specific skillset pertaining to aerodynamics course. The skills acquired by the students in the aerodynamics course are:

1. Using XFLR5 or XFOIL to perform basic aerodynamic simulation and predict coefficient of pressure and coefficient of lift distribution on an airfoil and a wing.
2. Numerically integrating (both line and area integral) using Matlab.
3. Perform symbolic integration and differentiation in Matlab along with solving polynomial equations.
4. Perform numerical integration and differentiation in Matlab.
5. Plotting streamlines, streaklines and pathlines for any given flowfield using Matlab.
6. Numerically determining vorticity and finding circulation for a given experimental data.
7. Determining lift from circulation of wingtip vortices using experimental data.
8. Combining multiple elementary flows to simulate flow over complex geometries in Matlab.
9. Determining velocity and pressure distribution along a surface of an object created through combination of elementary flows.
10. Designing series of airfoils through Joukowski Transformation in Matlab.

11. Simulating inviscid flow over airfoil designed Joukowski transformation in Matlab.
12. Theoretically determining the lift curve slope, zero lift angle of attack and center of pressure for the airfoil created using Joukowski transformation.
13. Coding source panel and vortex panel methods in Matlab to simulate flow over cylinder and airfoil.
14. Converting airfoil to a finite wing and running simulation in XFLR5.

Achieving most of these technical outcomes involve students learning to program in Matlab. Programming is one of the ways through which students can see a tangible outcome for concepts such as vorticity, circulation, elementary flow, etc. Without seeing the flows and experiencing the changes themselves, the students won't be able to realize the significance of the equations they learned. However, technical outcomes must be connected with the learning outcomes of the class to achieve higher stages of learning. This connection between the learning outcomes and the technical outcomes of the course is achieved using the portfolio mentioned earlier.

For example, for the technical outcome of using XFLR5 and XFOIL to perform basic aerodynamic simulation, in order to satisfy the set learning outcomes in the previous section, they need to:

1. State when and where this skill can be applied in real-life with specific examples.
2. Explain the process of simulation and the results obtained from simulation along with sanity check.
3. Connect how the class discussion on coefficient of pressure and coefficient of lift ties into the result obtained from simulation.

These three questions satisfy the first three learning outcomes listed in the previous section for one technical outcome of the class. Similarly, all the technical outcomes can be corresponded with the learning outcomes to achieve deep or higher stages of learning and understanding. Usually, students connect the teaching outcomes and the learning outcomes in the reflection portion of the portfolio. Designing the class around these learning objectives where technical outcomes are combined with team work, communication and critical thinking skills have made a monumental difference in students' learning.

*"When I explain to people that I do not have tests and quizzes, they look at me like I am overthinking a relatively easy course. However, what is demanded for the homework and the homework revisions, I know the material better than I would with any tests or quizzes, because I have had to explain it, multiple times! Additionally, the skills we developed in XFLR5, in the wind tunnel and in MatLab will be with me for a very long time mainly because I learned through reflecting. I really appreciated this class, and even though it was a lot of work, it made me passionate about the subject!"*

*- Heather L*

The modules, practices and tools used to achieve the technical and learning outcomes are listed in the section below.

## 5. Modules and Practices

In order to achieve the set technical and learning outcomes, the following modules and practices were practiced in the Fundamental Aerodynamics class.

### 5.1 No "required" course textbook

While having a required textbook can be beneficial, for a subject as broad as Aerodynamics, it is not fair to have the students confined to just one textbook. Students should learn to find information from multiple resources through their own efforts. The modified syllabus contains a list of recommended

textbooks but the students were required to do independent studies from online resources. The alternate intention of not having a required textbook is to break free from the tradition of asking students to solve back of the chapter homework problems. Students were given open-ended problems which related the concepts they learned in class to the outside world. And the students appreciated the nature of the Homework problems that were given in the course.

*“I really liked the way the homework maps exactly to what we are doing in class because it makes you learn the material we are learning in class the right way. I have always considered myself to be a learner by doing and finally having a class in college where I am allowed to implement the things I learn in the class room is very exciting and exactly what I looked forward to getting out of this class at the beginning of the semester.”*

- Chris B

*“I think having the homework assignmentss be as challenging as they were, helped me to study and retain the material more than I would have for an exam. Overall, I enjoyed the course and really hope I can beneficially utilize a good deal of this information in the future.”*

- Austin O

*“I really like how you gave us real data to analyze in our homework; it was really cool to listen to the guest speaker talk about panel methods see that we analyzed and understood things he did research on. I also noticed this mostly with the plunging plate data. I have very much enjoyed this class, and like the portfolio style homework that we do. I find myself reminded of aerodynamics in everyday life, whereas before this class, I wasn't. This class has taught me a lot and I know I will use what I have learned in the future.”*

– Sarah G

Below are some of the resources which are recommended to the students through the syllabus.

#### **5.1.1 Books:**

**Recommended Book:** Fundamentals of Aerodynamics, J. D. Anderson

One of the most recommended textbooks for anyone who is starting in the field of aerospace engineering is Anderson's Fundamentals of Aerodynamics. The textbook is well written in a way that any layman can understand the concepts and the equations. It is for this reason most instructors who teaches aerodynamics default to using Fundamentals of Aerodynamics by Anderson as a required textbook. While the explanations in Anderson textbook are easy to understand, it doesn't quite provide avenues to let the students critically think about a concept. And there are some concepts such as Joukowski transformation which are not covered in the Anderson textbook. So students were encouraged not to stick to this book and were encouraged to refer multiple books to gather resources and knowledge.

**Reference Books:** Foundations of Aerodynamics, A.M. Kuethe, J.D. Schetzer, C.Y. Chow  
Aerodynamics for Engineering Students, E. L. Houghton, P.W. Carpenter  
Aerodynamics for Engineers, John J. Bertin, Russell M. Cummings

Some of the topics discussed in class were also taken from the books mentioned above. All these books should be available in the library.

#### **5.1.2 Online Resources:**

Inquiry based learning is one of the major learning outcome of this class. As such, students are required to do online research on topics discussed in the class. There are many online resources (mentioned below) which helped students learn more about the subject.

MIT Open Course - <https://ocw.mit.edu/courses/aeronautics-and-astronautics/16-100-aerodynamics-fall-2005/>

NPTEL Open Course - <http://nptel.ac.in/courses/101105059/>

History of Aerospace Engineering - <http://theageofaerospace.com>

### 5.1.3 Android apps:

To make use of all the technology, the students were exposed to the android and OS apps. Below are QR codes for some fun and useful mobile apps related to Aerodynamics.

#### **Aerodynamics Tool Free**



<https://goo.gl/BFAJ9K>

#### **WindTunnel Free**



<https://goo.gl/OMxT5H>

iOS version available at <http://www.algorizk.com/>

Most of what mentioned in this section was included in the syllabus of the aerodynamics course. Including these in the syllabus proved to be effective in communicating to the students that the aerodynamics course won't be taught using conventional lecture based approach.

## 5.2 Guided notes

As mentioned before, fundamental aerodynamics course in nature consists of multitude of equations and derivations. The course was taught by the author in the evening (6.30 PM) twice a week. Due to the timing of the class, the traditional approach to deriving the equations on the white board and having them copy in their notes will result in boredom and awkward silences among students. As such, a new approach is needed to perform the derivations in class which engages students as well as makes them critically think about the equations involved. Once such method is the use of Guided notes.

Guided notes are instructor-prepared handouts provided to all students in the class with necessary background information and cues with unfilled spaces to write key facts, equations, concepts, assumptions, etc during the lecture. An example of guided notes is included in the appendix for the Thin Airfoil Theory, one of the most famous theories in the realm of aerodynamics. Some of the important assumptions, derivation steps, and equations were intentionally left out from the notes so that the students can discuss in groups and fill in the blanks. Through this method, the students were able to pay complete attention to the class and were able to ask important questions to the author. The guided notes surprisingly provided a faster way to go through the derivations when compared to writing equations on the board. When it comes to writing on the board, the author particularly does not have a good handwriting which leads to misinterpreting some of the mathematical symbols and parameters by the students. The guided notes provided a way for the students to accurately document the derivations as well. This method can be applied even for a larger class. The instructor should walk around the class and make sure that students are working on the guided notes without being distracted. And students seemed to like the guided notes better than copying from the board.

*“Throughout the course, I found the guided notes always seemed to really help me retain information and I also think the homeworks, although sometimes lengthy, did force me to think more deeply about the topics that were covered in order to gain a better understanding of them”* – Austin O

The author is currently teaching the same aerodynamics course where guided notes are used. Figure 5 shows the picture of the classroom with students working on the derivation using guided notes. By writing the derivations on the board, the students explain the steps to their peers and engage in conversations. The classroom atmosphere is lively while students are deriving the equations. Such a scene is not possible in a classroom where instructor derives the equations on the white board and the students copy it.



**Figure 5 Students working in groups deriving equations using guided notes in classroom**

### **5.3 Visual representation of concepts using slides**

The author found that visual aids are extremely beneficial for students to understand the concepts especially when they are complicated. Power point slides were used in each class to aid teaching of concepts. In most conventional classrooms, teaching using power point involves instructor reading from the notes and students listening to it. The main focus of the class is the power point slides which is not conducive to learning. Instead, the author used PowerPoint slides to show multiple animations, figures, contours, graphs, etc which aids in students’ understanding of the subject. This tool proved to be extremely useful when the author taught the concept of source panel method. Source panel method can be considered as one of the complex topics in Fundamental Aerodynamics by the students. It took the author three classes to cover just one topic and it couldn’t have been done without the use of slides. And students found the animations and cartoons useful as seen in Table 2.

*“My favorite was the day we did the source-panel method and it clicked that by finding influence of sources on each other, we can calculate the velocity. I think everyone else knew it the entire time. However, for me as the words left my mouth it felt like I had come up with the idea myself and I was struck by how ingenious of a method it was.”* – Eric I

Table 2 Likert scale Feedback on Panel Methods Lectures and Homework Assignments

| Questions                                                      | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|----------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| Lectures on panel methods were interesting                     |                   |          | 5       | 15    | 2              |
| I understand the significance of panel methods can do          |                   |          | 2       | 10    | 10             |
| The powerpoint presentations on the panel methods were helpful |                   |          |         | 17    | 5              |
| Panel methods was hard to understand                           |                   |          | 6       | 12    | 4              |
| Coding in class was helpful                                    |                   |          | 2       | 4     | 16             |
| The delivery of panel method concept was clear                 |                   |          | 4       | 18    |                |
| I still don't understand how panel method works                | 1                 | 15       | 5       | 1     |                |
|                                                                | 1                 | 15       | 24      | 77    | 37             |

## 5.4 Surprise quizzes

The author gave two surprises quizzes in the spring semester when the course was taught. The first surprise quiz was given in the first class after the spring break and the second was given in the first class after the Easter break. The intention behind the surprise quiz is to test the knowledge retained by the students WITHOUT ANY PRIOR PREPARATION.

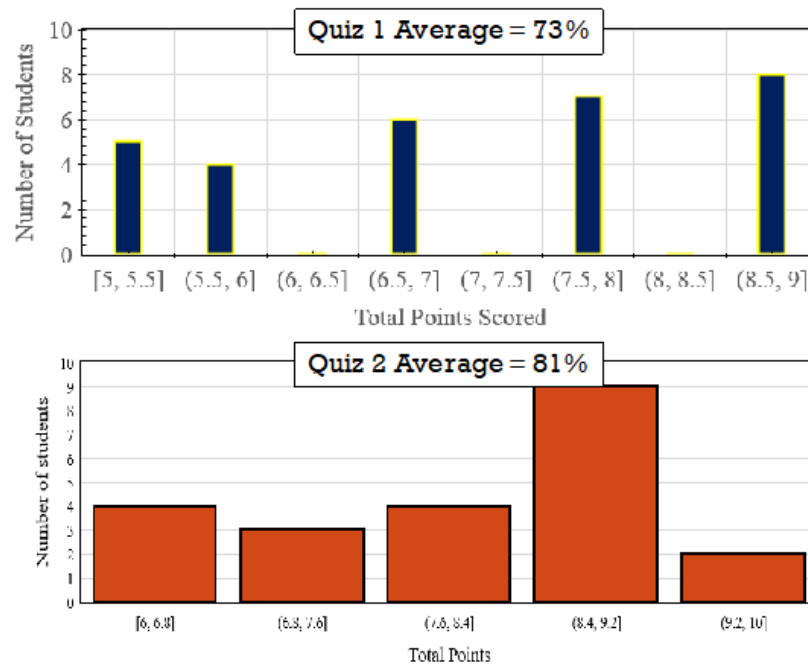


Figure 6 Surprise Quiz Results

Knowledge is a byproduct of understanding. Even though understanding is at the bottom stage of learning in the Bloom's taxonomy, it is not easy to achieve in a classroom. If the students really understood a topic or a concept, the necessity for preparation diminishes. The surprise quizzes after the spring break was a way to assess deeper understanding of the students as well as the effectiveness of the teaching pedagogy

and the modules used in the course. The results from two quizzes are shown in Figure 6. The average of the first quiz was 70% and the average of the second quiz was 80%. The scores indicate that the teaching pedagogy employed in the course was effective in achieving deep learning among students and the effectiveness increased as the students became more tuned with the course style and structure. The two quizzes which were handed out to the students are included in the appendix.

## 5.5 Student and Instructor Reflections

As a part of portfolio, students are required to reflect on the things they learn in class, in homework, in projects and other elements of the course and mentioned in the previous sections. And most students especially engineering students struggle providing reflections because they are not used to it and most engineering classes doesn't require the students to reflect on the things they do. As such requiring the students to do reflections in the aerodynamics courses was met with lots of questions about reflections and the content that goes into the reflection. The author gave the following questions to be used for reflections:

- What are the most important things you learned in this topic?
- What questions still remain on this topic?
- What steps will you take to try to answer these questions – may lead to an independent research topic, if so, please make reference to this?
- How can you use the information presented in your daily work?
- What real life examples from your life/other classes/experiences can you share with the class?
- Do you have any suggestions for the professor on how to better present this topic?

Some of these questions are tailored to connect the technical outcomes and the learning outcomes. The author strongly believes that students respond better when they see instructors doing the work that they were asked to do. Therefore, the author started writing reflections in an online communication tool used at the University of Dayton after each class. As an instructor, providing reflections after each class was extremely beneficial. In the reflection the author talked about the topics discussed in class, the style of teaching, the homework, etc. It gave the author another platform to reach to the students. The author stated the intention and purpose behind the teaching style, homework and projects so that the students knew without a doubt what they are doing and more importantly WHY! It proved to be an effective way to involve students as the we navigate through the class especially the struggles in preparing for a certain topic or a concept. And example of instructor reflection is shown below.

“Hey guys,

*I finally had time to sort through the Likert scale assessment despite multiple advising sessions all day. I wanted to wait before I write this because I need to reflect on the Likert scale assessment as well. Lets start with the class. What do you think about the last class? Do you think the way the thin airfoil theory was introduced makes sense? Were you able to follow the different steps involved in the derivation of the theory? I am not a big fan of deriving equations because it doesn't add much value in the application. But at the same time, we need to know where the equation we use on a daily basis come from and how it was derived and all the assumptions made to get to the result. You just need to get exposed to it once. And thin airfoil theory is a universally accepted theory which compares extremely well with the experimental result. All thin airfoil theory does is it mathematically models vorticity distribution as seen in the boundary layer in real life to find lift force and lift coefficient. The way it does that is by finding the magnitude of vorticity distribution that makes the camber line the streamline of the flow and then finding lift force/coefficient from the circulation which is just the integration of vorticity.*

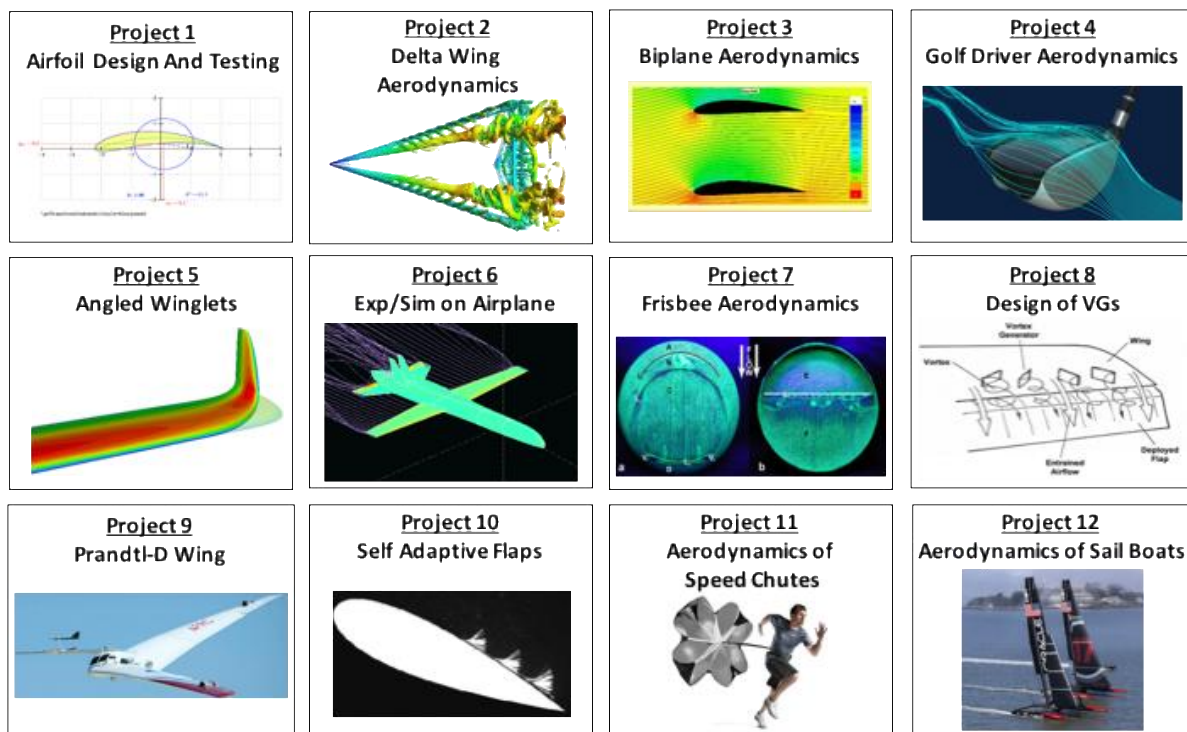
*I hope the guided notes were useful in helping you step through the derivation. I will post the word file of the guided notes online as well so that you can use the equations in your homework. The equations*

are also one the presentation slides. My goal is to try to take the focus away from the equations and put it on the application and practice. That is the intention behind creating passion projects and homework assignments. I don't care about how you perform in a one-hour/two-hour exam. That's why we don't have exams. I care about how you perform on the five learning objectives which I mentioned at the start of the class.”

- Sid G

## 5.6 Passion Projects

The passion projects were one of the most important module of the course. The idea of passion projects was inspired based on a project-based research class for undergraduate students at Massachusetts Institute of Technology.



**Figure 7 Examples of passion projects that students undertook in Fundamental Aerodynamics class**

When students think back on their formidable experiences in the university, it won't be the lectures or the classes they take but the opportunity to take ownership over an independent project and pursue a single idea. This experience will cement student's love for learning and the process of discovery. This was the inherent motivation behind designing the passion projects because it promotes independent study, sustainable and residential learning. It is designed so that students acquire the skillset of scientific investigation through independent research, testing and analysis. Throughout the semester in the fundamental aerodynamics class, students had the option to choose two different passion projects of their interest out of 15 projects or come up with their own project. The students got a chance to teach their peers about what they learned and enhance the quality of conversation and learning in the classroom. Some of the passion projects that the students worked on is shown in Figure 7. As evident from the figure, the students tested multiple application of aerodynamics and not just pertaining to airplanes. The author created a video compiling the passion projects done by the students. The video can be found here - [goo.gl/1vJwGF](https://goo.gl/1vJwGF). Each student in the class did two passion projects with different applications of aerodynamics.

Even though the passion projects are open ended, without proper scaffolding is required for efficient project management. The steps that each student undertook in the passion projects are:

1. Topic selection
2. Literature review
3. Identifying dependent and independent variables of interest
4. Formulation of test matrix
5. Gathering resources to test in the wind tunnel
6. Experimental setup
7. Performing experiments
8. Data analysis
9. Discussion and Reflections

As stated in the sections earlier, the passion projects are incorporated into the homework so that students can find connections between their project and the topics that they learned in class and explain the connection in their portfolio. Apart from the experience, the skills the students' learned from passion project are:

1. Doing independent research/literature review on a topic they are passionate about.
2. Determining the deliverables for certain applications (what to change and what to measure).
3. Finding resources for the test setup/simulation.
4. Estimating order of magnitude of forces in a given test model so that it won't exceed the range of the sensor.
5. Putting together a test setup based on the available equipment.
6. Performing basic aerodynamic testing.
7. Data analysis with sanity check.

The feedback from the students on the passion projects were positive as seen in Likert Scale assessments shown in Table 3 and Table 4. For some students, the passion projects provided a great platform for getting started with research.

*"Another highlight for this class for me were the passion projects. I had a lot of fun working in the wind tunnel being able to test and apply what we learned in class. The second project we did, creating Joukowski Airfoils was my favorite of the two. Just learning about the Joukowski transformation was fascinating, and I enjoyed being able to apply it even more. I had a role in testing and picking which transformation we would use for three of the airfoils. Although we had many struggles with the wind tunnel gremlins I enjoyed my time experimenting. It was a very nice way to wrap up the semester."*

- Jen C

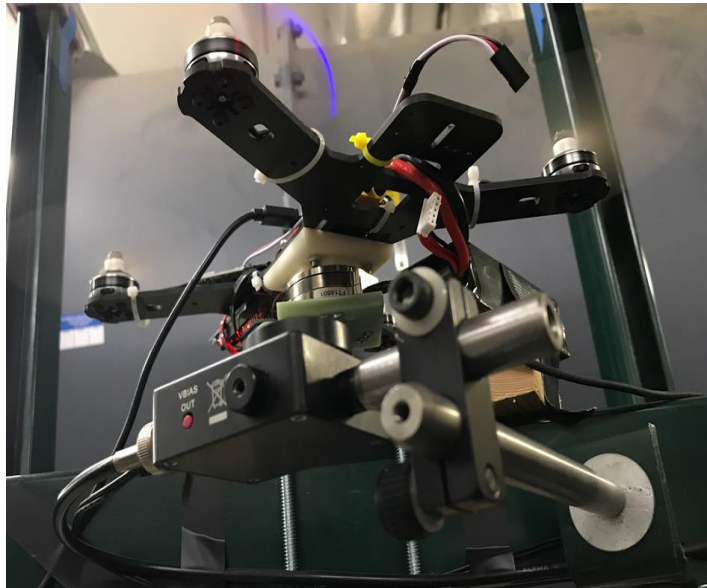
*"The Passion Projects were the first time I've gotten to research and test something on that scale and of my own interest. Though the math was difficult, I know this is the class that I will remember the most from conceptually. Coming into UD I didn't know where I wanted to take my Mechanical Engineering Degree, however through my journey I have learned that I am extremely interested in aerospace and this class just increased my desire to go into aerospace engineering. Though it is going to be extremely difficult, I hope to eventually get my masters in this field. Thank you for the great semester."*

- Eric I

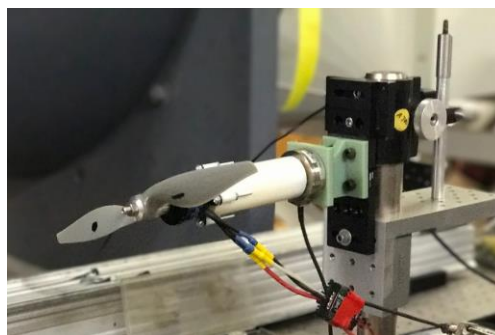
Some of the pictures of passion projects that the students conducted at the University of Dayton Low Speed Wind Tunnel is shown below.



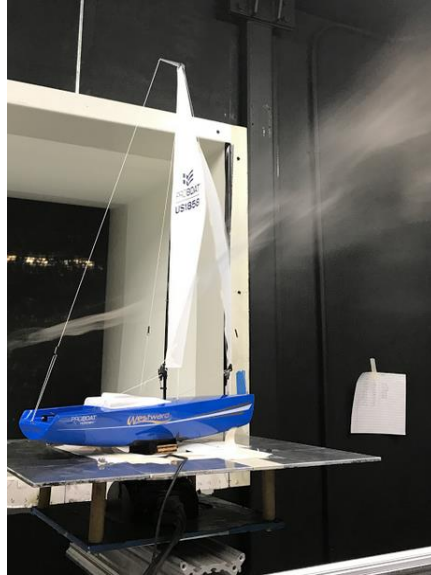
**Figure 8 Aerodynamic testing of speed chute to determine the additional resistance gained by athletes.**



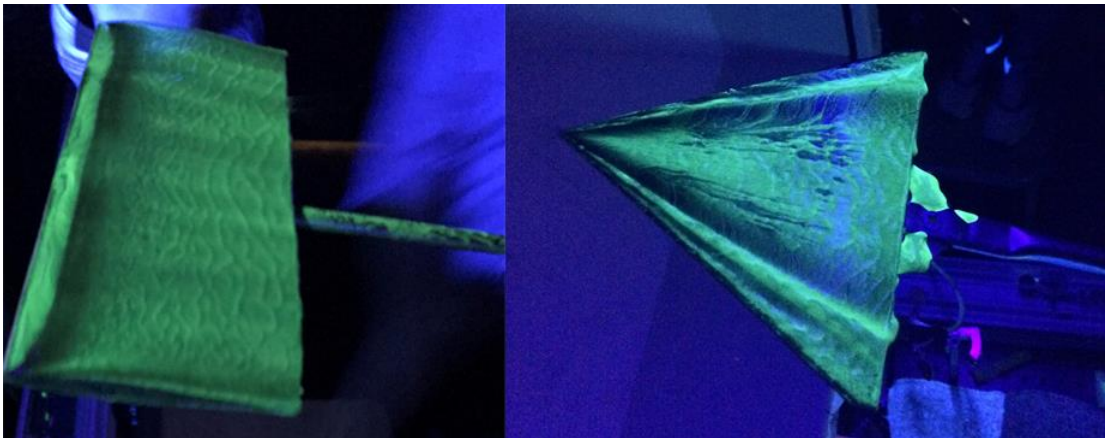
**Figure 9 Performance testing of in-house built quadcopter through force and moment measurements.**



**Figure 10 Testing of propellers to determine changes in performance due to ground-proximity**



**Figure 11** Smoke visualization and force-based testing of sail boat to determine optimum sail angle with respect to the wind.



**Figure 12** Oil flow visualizations and force measurements on rectangular wing and delta wings. The leading edge separation bubble along can be clearly seen on the upper surface of the rectangular wing. In the delta wing, the apex vortices generated at the leading edge can clearly be seen as well.



**Figure 13** Measuring drag of a golf ball using momentum deficit principle

## 5.7 Multiple Likert Scale Assessments

Students respond better if they are aware that the instructor values their feedback and is capable of tailoring the teaching technique to improve the classroom atmosphere and overall progression of the course. Therefore, multiple Likert scale assessments were taken during the course of the semester to assess students' thoughts on the modules and teaching styles practiced in the class. Two such Likert scale assessments are mentioned in Table 3 and Table 4. The Likert scale assessment shown in Table 3 was taken in the middle of the semester and the assessment shown in Table 4 is taken at the end of the semester. The Likert scale assessment contains questions related to the modules and practices mentioned in the previous sections as well as the overall teaching pedagogy. In the middle of semester, most students found the subject matter to be interesting but almost all students found the subject matter to be interesting by the end of the semester. This provides evidence that the teaching pedagogy used in this course improved students interest and passion towards the subject. Almost all the students loved using the guided notes. Most students also appreciated the instructor reflections as well.

Table 3 Likert scale assessment taken mid semester

| Questions                                                            | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|----------------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| The class structure is conducive to learning                         |                   | 1        | 3       | 11    | 12             |
| The teaching style makes me be attentive in class                    |                   |          | 4       | 12    | 11             |
| Subject matter is interesting                                        |                   |          | 1       | 14    | 12             |
| The powerpoint presentations are helpful                             |                   | 1        | 1       | 8     | 17             |
| Guided notes are helpful                                             |                   | 1        | 1       | 5     | 20             |
| Homework problems are helping to apply the concepts learned in class |                   | 1        | 4       | 17    | 5              |
| Homework problems are helping to learn the subject                   |                   | 2        | 5       | 13    | 7              |
| I learned a lot in Passion Project 1                                 | 1                 |          | 5       | 14    | 7              |
| Instructor feedback on HWs are helpful                               |                   |          | 2       | 11    | 14             |
| Instructor reflections on each class is helpful in some way          |                   | 2        | 3       | 15    | 7              |
|                                                                      | 1                 | 8        | 29      | 120   | 112            |

Table 4 Likert Scale Assessment taken at the end of the semester

| Questions                                                             | Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree |
|-----------------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| The class structure was conducive to learning                         |                   |          | 2       | 11    | 13             |
| The teaching style made me be attentive in class                      |                   |          | 1       | 15    | 10             |
| Subject matter was interesting                                        |                   |          | 1       | 8     | 17             |
| The powerpoint presentations were helpful                             |                   |          |         | 6     | 20             |
| Guided notes were helpful                                             |                   |          | 1       | 5     | 20             |
| Homework problems were helping to apply the concepts learned in class |                   |          |         | 15    | 11             |
| Homework problems helped to learn the subject                         |                   |          |         | 15    | 11             |
| I learned a lot in both passion projects                              |                   | 1        | 1       | 8     | 16             |
| Instructor feedback on HWs were helpful                               |                   | 1        |         | 9     | 16             |
| Instructor reflections on each class was helpful in some way          |                   |          | 3       | 11    | 12             |
|                                                                       | 0                 | 2        | 9       | 103   | 146            |

## 6. Conclusions

The integrated teaching model was practiced for the second time in an undergraduate Aerodynamics course. The teaching model was modified to incorporate both instructor-centered and student-centered learning which lead to deeper understanding and learning among students. Higher levels of learning in the classroom was also achieved by effectively combining learning outcomes with technical outcomes of the course. Portfolio was used as a platform to help students make the connection between the technical outcomes and the learning outcomes of the course through summary and reflections. Multiple teaching modules and practices were employed inside and outside the classroom to achieve the set technical and learning outcomes of the course such as no required textbook, using guided notes, providing instructor reflections, surprise quizzes, passion projects and taking multiple Likert scale assessments. Some of the logistics of the aerodynamics course is mentioned below:

1. 28 Classes
2. 28 Instructor Reflections
3. 12 Homework Assignments
4. 1 Guest Lecture
5. 2 Surprise Quizzes
6. 15 Handouts/Guided notes
7. Numerous group discussions
8. 2 Passion projects

The results from the assessment of the modules indicate that the overall interest in the subject matter was increased among the students and they learned a lot in the course. Likert scale assessment also indicated that the students were able to achieve the set learning outcomes of the course. While the model had a great

impact on the students' learning and understanding of the material, the workload of the students is comparatively greater when compared to other classes.

## 7. References

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- [9] Collis, K.F. and Biggs, J. (1986). "Using the SOLO taxonomy". Set: Research Information for Teachers, 2(4).
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- [11] Keen - The Framework <https://engineeringunleashed.com/Mindset-Matters/Framework.aspx>

## APPENDIX

### Homework 1

Due: Tuesday 1/24/2017

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Dear students,

Once again, welcome to the Fundamental Aerodynamics class. The idea behind this homework is to get you start writing your portfolio and start thinking about the subject of aerodynamics. It also gives me an idea of what you already know about the subject. This homework is intentionally abstract and open-ended as advertised in the syllabus. Please be descriptive in your answers as opposed to one line/one sentence answers. More often than not, you will get questions like these along with solving numerical open-ended problems. I strongly believe that by practicing this kind of problems, you will become better thinkers and learners. Always remember that for numerical or abstract problems, I don't care much about you getting the "right" answer. I care more about the approach you take to solve a problem. So focus on the approach. **You can turn in the homework via Isidore.**

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#### **1. Title Page**

Please begin your portfolio with a title page. (You can have a content page. However, it is optional.)

#### **2. Setting Big Picture Context**

Before you start your portfolio, it is very important to set a big picture context as to why we are doing this. Please provide a brief reflection on (not necessarily in order)

1. Your goals, aspirations and your interest in the field of aerospace engineering
2. Why are you taking this class in the first place?
3. What are you hoping to learn from this class?
4. How do you think this class helps you in your academic and professional career?

*(You don't have to retype these questions in your portfolio. Your narrative should have answers to these questions.)*

#### **3. Questions to Answer in Portfolio (50 Points)**

*(The three questions posed below, will make a good introduction to the subject of aerodynamics in your portfolio. Once again, you don't have to answer these questions in order. Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc.)*

1. Select three different applications of aerodynamics we discussed in class. Based on what you know and what you are aware of, provide a brief description of the role of aerodynamics in each application. Think about the type of flows experienced by those applications and list the dominant flow parameters (Ex: Pressure, Friction, Lift, etc,) acting on those objects.
2. What do you think are the similarities and differences among the aerodynamics of insects (Ex: dragonflies), birds, airplanes, and rockets? Think about the mechanisms used to generate lift, drag and thrust forces in these applications and talk about how they generate those forces.
3. You will be using a lot of math in class and it is probably good to refresh on some of the basic concepts. Explain *\*clearly\** what **dot product**, **cross product**, **divergence**, and **gradient** mean when applied to a scalar and a vector. **I AM NOT LOOKING FOR JUST EQUATIONS. YOU HAVE TO EXPLAIN WHAT THEY MEAN IN PHYSICAL SPACE.** Also, provide one application for each where you use/apply those operators.

#### **4. Provide Global Reflections on the First Week of Classes and Homework 1 (15 Points)**

# Homework 2

Due: Tuesday 1/31/2017

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Dear students,

This homework should get you to think about calculating coefficient of lift, drag, and pressure. Mainly focusing on the coefficient of pressure. Pretty soon, you will be able to generate the  $C_p$  distribution by yourself. But before that, you should be comfortable dealing with pressure coefficient and understand the significance of it. That's what this homework is about. We are building tools and skills required so that we can move forward with actual simulation.

---

## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc.)*

1. To continue from where you left off from the previous homework, state briefly why should we care about aerodynamic coefficients? You don't have to explain each and every coefficients in detail but you have to provide evidence to show WHY coefficients are important.

2. We discussed in class that just by looking at the variation of Coefficient of pressure, you can tell a lot about the aerodynamics around any object. Give a brief explanation on coefficient of lift. As an example, please look at the Coefficient of Pressure contours of a D-8 double bubble design and the pressure coefficient of blended wing body in the power point slide. Based on the pressure coefficient, explain the aerodynamics around these configurations. In your opinion, which of these configurations produce better aerodynamic efficiency?

3. Please see attached pamphlet on the CGS Hawk Light Weight Sport Airplane. If you read the content, you will see that the airplane uses semi-symmetrical airfoil when compared to flat-bottom airfoil used in other light-weight sport airplanes.

1. Run an XFOIL analysis (tutorial given in slide) to determine the pressure distribution on a semi-symmetrical NACA 2412 airfoil and a flat bottom Clark Y airfoil at the same angle of attack and at the cruise Reynolds number (consider sea level). Compare the coefficient of pressure distribution between the two airfoils and find out what advantages you get by having a semi-symmetrical airfoil over a flat-bottom airfoil.

2. According to the given specifications on the sheet, what would be the velocity at which the airplane will have the longest range? Clearly state the rationale behind the approach you took to solve the problem. Perform "sanity" check on the answer you get.

**(Fun fact:** The entire airplane costs around \$25000. And with a 15 mph head wind, the airplane could stand still in air. The pilot even said he was able to fly backwards which stunned some of the F-16 pilots!)

## **FOR GRADUATE STUDENTS ONLY**

4. Consider a circular cylinder in a hypersonic flow, with its axis perpendicular to the flow. Let  $\phi$  be the angle measured between radii drawn to the leading edge (the stagnation point) and to any arbitrary point on the cylinder. The pressure coefficient distribution along the cylindrical surface is given by  $C_p = 2 \cos^2 \phi$  for  $0 \leq \phi \leq \pi/2$  and  $3\pi/2 \leq \phi \leq 2\pi$  and  $C_p = 0$  for  $\pi/2 \leq \phi \leq 3\pi/2$ . Calculate the drag coefficient for the cylinder, based on the projected frontal area of the cylinder.

## **2. Provide Global Reflections on the Second Week of Classes and Homework 2 (15 Points)**

# CGS Hawk

Specifications and Pricing

# SPORT

[www.cgsaviation.com](http://www.cgsaviation.com)

1-863-430-5829

## Hirth Engine 3202-55 HP



### Specifications

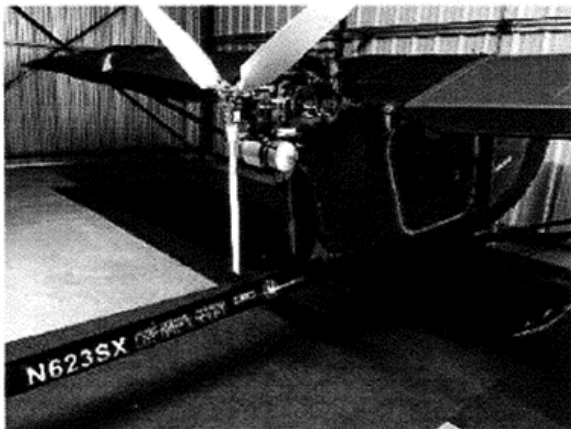
|                        |              |
|------------------------|--------------|
| Empty Weight           | 310 lbs      |
| Gross Weight           | 625 lbs      |
| Wing Span              | 28' 10"      |
| Wing Area              | 135 sq. ft   |
| Engine Size            | 40-65 hp     |
| Rate of Climb          | 800-1100 fpm |
| Cruise Speed           | 55-75 mph    |
| Stall Speed            | 28-34 mph    |
| Vne                    | 100 mph      |
| Assembly Time          |              |
| For first time builder | 200 hrs      |

### CGS Hawk "Sport" Single Seat

### *\*Powerfin Props\**

The Hawk "Sport" Single Seat is the "Hot Rod" version of the standard Classic Single Seat. It was first developed to compete in the European World Air Games in 1999. This competition is held once every two years in different countries. In the last three competitions at the World Games, the Hawk Sport and it's owners have won First Place (two competitions in a row), Second Place, and a Third Place.

The Hawk Sport differs from a Classic in several ways. First, It has a double surface wing with a semi symmetrical airfoil instead of a flat bottomed one like the standard Classic. This is achieved by the addition of lower rib battens on the lower surface of the wing forming the airfoil. This allows the stall speed to remain the same but the top speed to increase. The wing struts are a fully streamlined aluminum extrusion, which doesn't require the use of jury struts. This allows for less drag and more speed, or you can cruise at the same speed with a lower power setting. In addition, the gear legs have streamline fairings over the fiberglass gear legs for even more drag reduction. The Sport also incorporates a curved overhead and curved windshield as standard equipment. The curved overhead offers taller pilots an additional 3 " of headroom; enough for tall pilots with helmets. A fiberglass nose cone also reduces drag and offers a sleeker appearance. The Hawk Sport model has the same top speed as the standard single seat Hawk Arrow.



# Homework 3

Due: Tuesday 2/7/2017

Dear students,

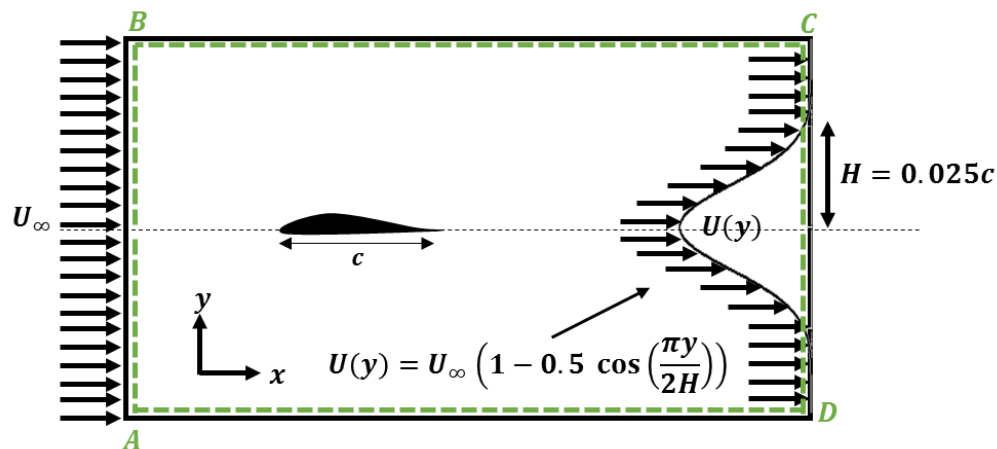
This homework should get you started on documenting the literature review for the passion projects. We will be discussing equations of fluid motion in class today and this homework will give you some experience in dealing with the equations of fluid motion. These equations are universal and can be applied to almost any fluid system. You should have some background to these equations from your fluid dynamics class. This homework will be a good review for you to apply some of the equations, which can give you important aerodynamic parameters.

## **1. Questions to Answer in Portfolio (70 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc.)*

1. First step in any experimentation is doing a literature review because it will give you a broad background on the application you are looking at. Document the relevant information you gathered from different resources you reviewed for your passion project 1. Based on your literature review and independent study, discuss the relevant aerodynamics involved in the application you will be working on. You have to include relevant graphs, tables, figures, etc to support your narrative.

2. Momentum equation is one of the most powerful equation in fluid dynamics because applications of it are endless. Say you are tasked with finding the drag coefficient of a new airfoil design. You don't have a force balance to measure the drag force directly. But you found a velocity profile equation in the wake of a similar airfoil from the literature. Can you estimate the drag coefficient using that velocity profile equation? (State all the assumptions you make). And what instrument can you use to check the accuracy of the velocity profile in the wake of your new airfoil?



3. If an incompressible and inviscid fluid flow is free of heat transfer and shaft work, do we need to consider energy equation in our flow analysis?

## **2. Provide Global Reflections on the Second Week of Classes and Homework 3 (15 Points)**

# Homework 4

Due: Tuesday 2/14/2017

Dear students,

In the last homework, you practiced applying conservation of momentum to obtain drag coefficient. In this homework, you will understand the application of energy equation in incompressible and compressible flows and how energy is transferred between the different flow variables. This homework should also help you to start thinking about the test matrix for your experiments (if you haven't already). And for the first time since we started, you will use MATLAB to plot flowfields. That is awesome!

## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Provide any additional resources to your passion projects that you might have come across in the last week and tabulate your test matrix for your experimentation. List down the tools and equipment you need to do your experiment and your plans to obtain them. Please schedule an appointment with me (using google calendar or email) to discuss and finalize your test matrix and to brainstorm on the tools and equipment necessary to do the experiment.
2. Determine the temperature increase on the leading surfaces of Cessna 172 and C-130 during cruising and the Space Shuttle during reentry. Based on the values you obtain, do you think it is reasonable to exclude energy equation in incompressible flow and not in compressible flows?



Max Speed: 188 mph



Max Speed: 336 mph



Reentry Speed: 17,000 mph

3. Given the two-dimensional incompressible flow fields:

|                               |                    |
|-------------------------------|--------------------|
| a) $u(x, y) = x^2y$           | $v(x, y) = -xy^2$  |
| b) $u(x, y, t) = -xt$         | $v(x, y, t) = -yt$ |
| c) $u(x, y, t) = 1/(1 + t)^2$ | $v(x, y, t) = -t$  |

### **Undergraduate students:**

Pick any two velocity fields out of three and determine the equations for streamline, streakline and the pathline. For each flowfield, plot the streamline, streakline, and pathline on the same plot. (If you are ambitious, you can every try and animate the flowfield with respect to time. However, IT IS NOT REQUIRED.)

### **Graduate students:**

For all the velocity fields given, determine the equations for streamline, streakline and the pathline. For each flowfield, plot the streamline, streakline, and pathline on the same plot. (If you are ambitious, you can every try and animate the flowfield with respect to time. However, IT IS NOT REQUIRED.)

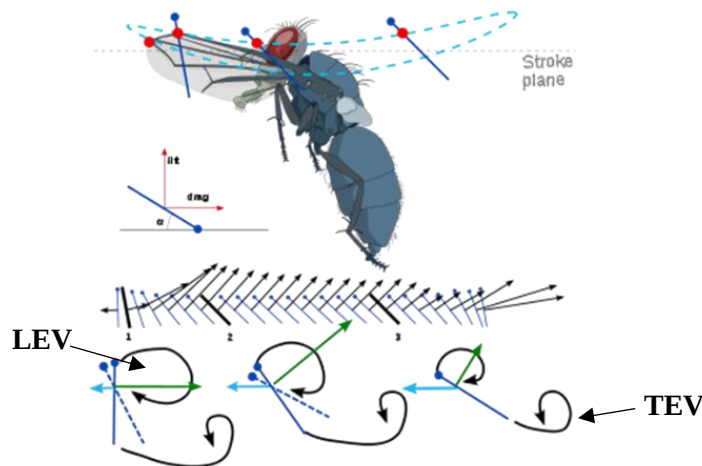
## **2. Provide Global Reflections on Week 4 and Homework 4 (15 Points)**

# Homework 5

Due: Tuesday 2/21/2017

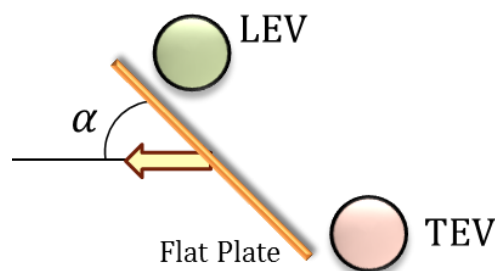
Dear students,

This homework will give you some experience in applying the aerodynamic tools such as vorticity and circulation in an actual research problem. Before we proceed, I need to set proper context here. When you look at the motion of an insect wing, it follows a complex pattern of pitching and plunging motions as shown in the figure below. Because of this, the aerodynamics of an insect's wing can be explored by studying the aerodynamics of pitching and plunging motions separately.



<http://www.scientificamerican.com/article/catching-the-wake>

When a wing plunges in air, it forms what is called as a Leading Edge Vortex (LEV) and a Trailing Edge Vortex (TEV). For this assignment, let us focus only on the LEV because it plays a monumental role in the production of lift on a plunging wing. To understand the formation and growth of this LEV, experiments were conducted at the Wright Patterson Air Force base in the Horizontal Free Surface Water Tunnel (HFWT) on a flat plate plunging at a certain velocity at a fixed angle of attack. Using an optical flow diagnostic technique called Particle Image Velocimetry (PIV), the velocity components around the leading edge of the flat plate was obtained. You can download the velocity vector file from Isidore.



The velocity vector file can either be opened in Excel or in Matlab. The file contains four columns – x, y, u and v. In this homework, you will numerically calculating vorticity and circulation. It is an essential skill to have when you graduate from this class because these concepts are applied on a day-to-day basis. There is no use of just knowing these concepts without applying. Since this assignment involves programming, **GET STARTED ON THIS ASSIGNMENT EARLY**. Don't wait till the weekend. Apart from finding vorticity and circulation, you also have to document any progress you made in your passion project in the last week.

## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Document any progress you made regarding your passion project such as finding resources, references and testing (if you have done it).
2. Plot the U velocity contour, V velocity contour and the velocity magnitude contour for the given experimental data and discuss the aerodynamics you see from the contours.
3. Determine the vorticity present in the flowfield and plot the vorticity contour. What new information regarding the relevant aerodynamics did you get by determining vorticity?
4. Explain the concept of circulation and its significance.

# Homework 6

Due: Tuesday 2/28/2017

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Dear students,

In the last homework, you numerically determined vorticity to see the leading edge vortex around a plunging plate. In this homework, you will use the same program you wrote to determine the vorticity and circulation of a wingtip vortex emerging from an aspect ratio 6 ClarkY wing! You will be able to see how the theoretical concept such as circulation compares to the experimental data. You will also be documenting the wind tunnel/simulation results from your passion projects in this homework as well. Similar to the last homework, don't postpone until the weekend to get started on this homework. Start right away.

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## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Hopefully, most of you finished your experiments in the wind tunnel. Please document the following:

- a) Order of magnitude estimate of the normal and axial forces (before you did the experiment)
- b) Experimental setup/model with pictures
- c) Results of your wind tunnel test. (Please don't include raw data. Plot the results in an appropriate way.)

**Note: You don't have to discuss your results in this homework. You will provide technical explanations for your results in the next homework. All you are doing in this homework is showing the results you got.**

2. Please download the ClarkY wingtip vortex experimental data files from Isidore under the "Homework" tab for different angles of attack.

- a) Use the program you wrote for the last homework to determine the vorticity of the wingtip vortex at different angles of attack. Identify regions of rotational flow and irrotational flow.
- b) Numerically calculate circulation in the wingtip vortex for each angle of attack.
- c) Use Kutta-Joukowski theorem to calculate lift and plot the lift as a function of the angle of attack.
- d) Compare the lift calculated from wingtip vortex to the lift measured from force balance. The force data is shown in the back of this page.

3. Consider the incompressible, irrotational, two-dimensional flow, where the stream function is given by,

$$\psi = 2xy$$

- a) What is the velocity at  $x = 1, y = 1$  and at  $x = 1, y = \frac{1}{2}$ . Do these two points lie along the same streamline?
- b) Graph the streamline pattern and discuss the significance of the spacing between the streamlines.

### **ONLY FOR GRADUATE STUDENTS**

- c) What is the velocity potential for this flow?
- d) Sketch the lines of constant potential. How do the lines of equipotential relate to the streamlines?

## **2. Provide Global Reflections on Week 6 and Homework 6 (15 Points)**

# Homework 7

Due: Tuesday 3/14/2017

Dear students,

Hope you all had a good spring break and came back rejuvenated and refreshed. We have been talking about simulating flows in this class for a while and in this homework, you are going to do just that. Using the stream function and Laplace equations, you will simulate flow over a cylinder and flow over a lifting cylinder. You will also be solving a problem related to a real life application using the equations we derived from potential flow around a cylinder.

## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Provide comprehensive technical discussions on the wind tunnel testing/simulation results you included in the last homework. You may consult with me if you could not figure out the science behind why you see what you see in your results.
2. Simulate the potential flow around a cylinder along with the velocity components. Show a plot of the stream function and a plot of the velocity vectors around the cylinder. Assume  $V_\infty = 1 \text{ m/s}$  and a doublet strength  $\kappa = -300 \text{ m}^3/\text{s}$ .
3. You are to design a Quonset hut to serve as temporary housing near the seashore. The Quonset hut may be considered to be a closed (no leaks) semicylinder, whose radius is 2.5 m, as shown in the figure below. Neglect viscous effects and assume that the flow field over the top of the hut is identical to the flow over the cylinder for  $0 \leq \theta \leq \pi$ . The air inside the hut is at rest and the pressure is equal to the stagnation pressure  $p_0$ . What is the force required to keep the hut on the ground if the wind speed is around 50 m/s? Consider sea-level conditions.



4. Simulate the potential flow around a spinning cylinder along with the velocity components. Show a plot of the stream function and a plot of the velocity vectors around the cylinder. Assume  $V_\infty = 1 \text{ m/s}$ , a doublet strength  $k = -300 \text{ m}^3/\text{s}$  and a vortex strength  $\Gamma = 100 \text{ m}^2/\text{s}$ . Determine the pressure distribution along the surface of the cylinder.

## **2. Provide Global Reflections on Week 7 and Homework 7 (15 Points)**

# Homework 8

**Due: Tuesday 3/21/2017**

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Dear students,

In the last homework, you simulated the flow over a stationary cylinder and a lifting cylinder. In this homework, we are going to kick it up a notch and simulate flow over an airfoil that you design. This is an extremely important skill to have. You can take this homework and carry it on as passion project as well if you are interested. You will design the airfoil using Joukowski transformation and simulate the flow using Joukowski transformation as well.

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## **1. Questions to Answer in Portfolio (75 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Compile the write-up of different sections of the Passion project 1 into one document. Please make sure you have the following sections in the final report

- a) Introduction
- b) Literature Review
- c) Order of Magnitude Estimate of Forces
- d) Experimental Setup
- e) Results and Discussion
- f) Conclusion

2. Please choose a second passion project to work out of the list of projects listed on the Class 16 powerpoint presentation. Please note that there are three projects added to the list of projects. You can also come up with a project you are interested in pursuing.

3. Use conformal mapping to simulate a flow around

- a) A symmetrical airfoil
- b) A cambered airfoil (**ONLY FOR GRADUATE STUDENTS**)

Make sure to use the circulation value which satisfies Kutta condition.

Show the following graphs:

- 1. Joukowski Airfoil
- 2. Flow around lifting cylinder
- 3. Flow around the shifted lifting cylinder
- 4. Flow around the airfoil

## **2. Provide Global Reflections on Week 8 and Homework 8 (15 Points)**

# Homework 9

Due: Tuesday 3/28/2017

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Dear students,

In the last homework, you simulated flow over airfoil and wrapped up Passion project 1. In this homework, you will apply the equations you learned in class on thin airfoil theory to predict the coefficient of lift, coefficient of pitching moment and location of center of pressure for a given camber line. You will also document the literature review on the project of your interest for the passion project 2.

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## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Document the relevant information you gathered from different resources you reviewed for your passion project 2. Based on your literature review and independent study, discuss the relevant aerodynamics involved in the application you will be working on. You have to include relevant graphs, tables, figures, etc to support your narrative.
2. Get X, Y coordinates for the cambered airfoil you generated from the last homework using Joukowski transformation. Input the coordinates into XFOIL or XFLR5 to
  - a) Determine the **coefficient of lift variation** with angle of attack and the **lift curve slope**.
  - b) Compare the result with **lift curve slope** predicted by thin airfoil theory and **zero lift angle of attack** predicted by thin airfoil theory.
  - c) Plot the **variation of the center of pressure** location with angle of attack.

## **2. Provide Global Reflections on Week 9 and Homework 9 (15 Points)**

# Homework 10

Due: Tuesday 4/4/2017

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Dear students,

In the last homework, you determined the coefficient of lift and the location of center of pressure as a function of angle of attack using XFLR5 for a cambered airfoil you generated. In this homework, you are going to program your code to determine the angle of attack of zero lift, coefficient of pitching moment and location of center of pressure based on the thin airfoil theory equations. You will be comparing the results you got with the results obtained from XFLR5.

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## **1. Questions to Answer in Portfolio (85 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. Provide any additional resources to your passion project 2 that you might have come across in the last week and tabulate the test matrix for your experimentation. List down the tools and equipment you need to do your experiment and your plans to obtain them. Please schedule an appointment with me (using google calendar or email) to discuss and finalize your test matrix and to brainstorm on the tools and equipment necessary to do the experiment.
2. Use the cambered airfoil you generated in the last homework and determine through thin airfoil theory,
  - d) The angle of attack at zero lift
  - e) Coefficient of pitching moment variation about the quarter chord with angle of attack
  - f) Variation of location of center of pressure with angle of attack

Compare these results with the results obtained from XFLR5 in the last homework.

### **Steps involved:**

1. Determine the camber line from the cambered airfoil you used in the last homework.
2. Use “polyfit” command to get the equation of the camber line. (Use 4<sup>th</sup> or 5<sup>th</sup> order polynomial fit)
3. Differentiate the equation of the camber line to get  $dz/dx$ .
4. Convert  $dz/dx$  in terms of polar coordinates using  $x = c/2 (1 - \cos \theta)$
5. Substitute in angle of attack at zero lift formula and integrate.
6. Compare the angle of attack at zero lift you get with the result from XFLR5.
7. Do similar procedure to calculate  $A_0$ ,  $A_1$  and  $A_2$  values you need to determine pitching moment coefficient and location of center of pressure.

## **2. Provide Global Reflections on Week 10 and Homework 10 (15 Points)**

# Homework 11

**Due: Tuesday 4/11/2017**

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Dear students,

In the last few homework, you created your own airfoil, simulated flow around that airfoil and determined lift coefficients, quarter chord and location of center of pressure using XFLR5 and thin airfoil theory analytical prediction. What is left is to achieve one of other goals where you can potentially get coefficient of pressure for any shaped object using panel methods (Basically what XFOIL and XFLR5 does). In this homework, you will simulate coefficient of pressure over an object using source panel method.

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## **1. Questions to Answer in Portfolio (50 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. For your passion project

- Document the order of magnitude estimate for forces/coefficients for your passion project 2
- Begin testing in the UD-LSWT

2. Download “incomplete\_sourcepanel.m” file from isidore. Fill in the missing steps and provide comments on the source panel code. (Hopefully, we will get time to do this in class)

Compare the coefficient of pressure obtained by source panel method with analytical distribution.

# Homework 12

**Due: Tuesday 4/27/2017**

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Dear students,

Here we are in the final assignment of the semester! It feels great to look back at the skills you acquired through these homework assignments. These assignments trained you to use and apply aerodynamic tools such as vorticity, circulation, KJ theorem and trained you to design your own airfoils using Joukowski transformation. You also did analysis on the airfoil you generated in XFLR5 and XFOIL and compared with theoretical predictions. In this homework, you will add on to your skillset by taking the airfoil you designed and converting it into a wing to perform XFLR5 analysis.

You will also be reflecting on a very important video by Albion Bowers on Prandtl-D. I could not think about a better way than to finish the theoretical aerodynamics class with the Prandtl-D wing. It is such a remarkable design which one day could fly in Mars! All that came from the theoretical aerodynamics.

Through these homework assignments, you documented several stages of progress regarding your passion project. I hope all of these assignments helped you understand the material and I hope the skills that you learned add value to your work/career wherever you may end up.

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## **1. Questions to Answer in Portfolio (50 Points)**

*(Your general narrative should contain answers to the following questions. Please be descriptive in your answers. Provide enough pictures, graphs, tables, etc. to support your narrative.)*

1. For your passion project 2, provide technical discussion and explanations for the results you generated in the wind tunnel/simulation. You may consult me regarding the results as well before you write it in your portfolio.
  
2. Model a 3D wing of AR 4 in XFLR5 using the Joukowski airfoil you created before. Perform viscous analysis on the wing using “lifting line” or “3D Panels” option in XFLR5 to determine
  1. Lift variation with angle of attack and lift curve slope. Compare lift curve slope with Helmbold’s equation.
  2. Induced drag variation as a function of angle of attack
  3. Span efficiency of the wing (Using Helmbold’s Equation)

XFLR5 Demo Videos –

<https://www.youtube.com/watch?v=P6AZTxZkojo&t=285s>

<https://www.youtube.com/watch?v=IFCL8IJYnI&t=153s>

3. Provide reflections (a minimum of 1 page with 12 point font and 1.5 in spacing) on the following video by Albion Bowers:

### **“The PRANDTL-D: Toward More Birdlike Flight”**

Link: <https://goo.gl/PSwmNp>

## **2. Provide Global Reflections on Week 12 and also On Fundamental Aerodynamics Class in General.**

# THIN AIRFOIL THEORY

## Guided Class Notes

Dear students,

In this handout, I am going to walk you through the derivation of one of the most important and famous theories in all the realms of aerodynamics – Thin Airfoil Theory. The result from this theory is surprisingly predicts with a high level of accuracy, the lift curve slope of any airfoil shapes. In this derivation, we will apply the thin airfoil theory to a symmetric airfoil and a cambered airfoil. This concept sets a good segue for panel methods. That is why I wanted to introduce this first before talking about the source and vortex panel methods.

As I mentioned in the slides, thin airfoil theory predicts lift by mathematical adding vorticity in the chord/camber line of the airfoil. This concept was derived from the real life because in real life, there will be vorticity in the boundary layer.

Consider Figure 1 where a vortex sheet of strength  $\gamma(s)$  is distributed along a camber line represented by  $z = z(x)$ .

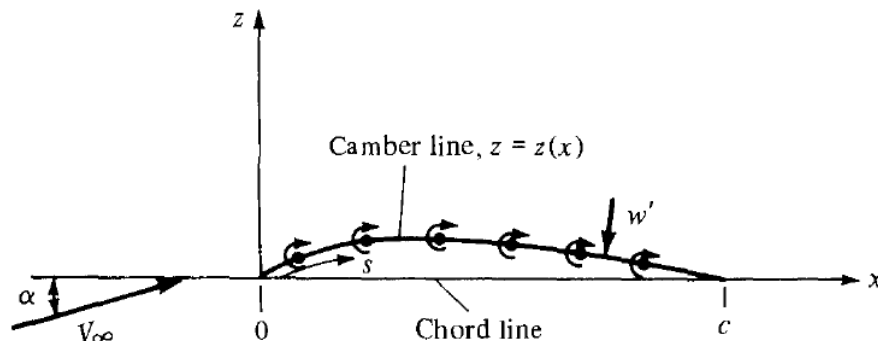


Figure 14 Vortex Sheet on a Camber line

In Figure 1,  $w'$  is the \_\_\_\_\_. Here we make a big assumption that the airfoil is thin. This will let us approximate the vortex sheet to lie on the chord line as shown in Figure 2.

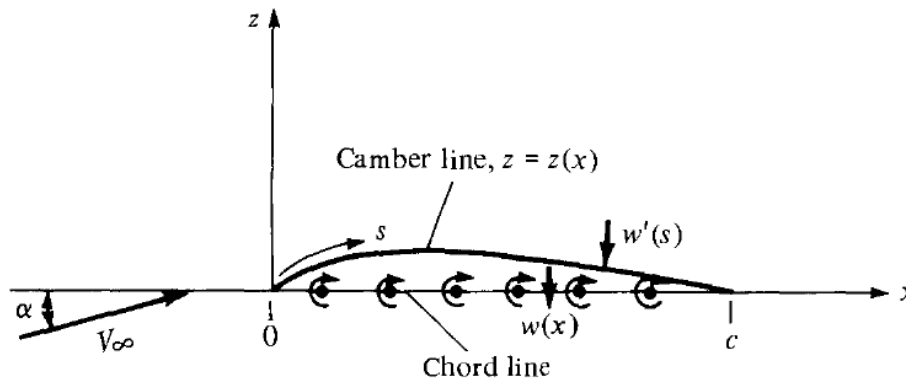


Figure 15 Vortex Sheet on a chord line

This also let us assume  $w'(x) = w'(s)$  which means the component of velocity induced by the vorticity sheet on the chord line is same as the component of velocity induced by the vorticity sheet on the camber line.

Now we impose a condition that

- 1.
- 2.

For the camber line to be a streamline of the flow, the component of velocity normal to the camber line must be zero at all points along the camber line.

Therefore,

$$V_{\infty,n} + w'(x) = 0 \quad (1)$$

Now, let us find an expression for  $V_{\infty,n}$  using Figure 3.

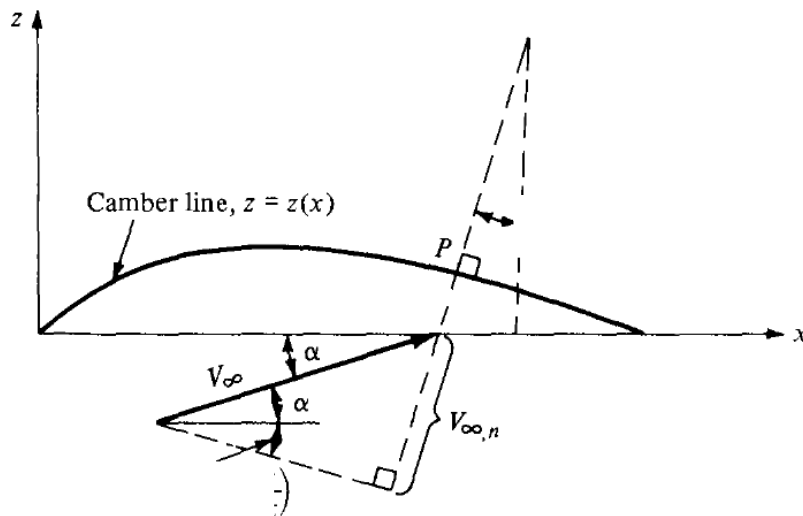


Figure 16 Determination of the component of velocity normal to the camber line

From Figure 3,

$$V_{\infty,n} = \quad (2)$$

Using small angle assumption, Equation 2 simplifies to,

$$V_{\infty,n} = \quad (3)$$

Now that we found an expression for the normal component of velocity in Equation 1, we need to find an expression for induced velocity as well. For that, please consider Figure 3,

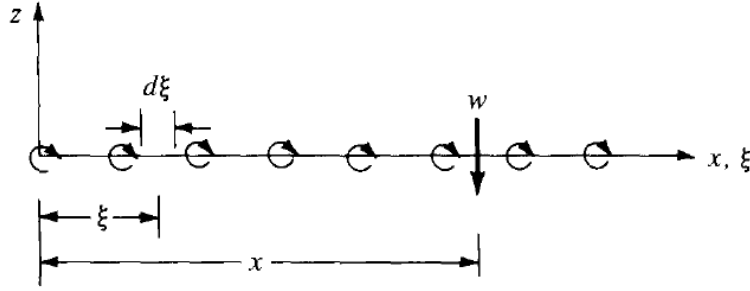


Figure 17 Calculation of Induced Velocity at the chord line

The induced velocity due to the vortex sheet can be written as,

$$dw = \quad (4)$$

The total induced velocity can be found by integrating Equation 4,

$$w' = \quad (5)$$

Substituting Equation 5 and Equation 3 in Equation 1,

$$V_{\infty} \left( \alpha - \frac{dz}{dx} \right) - \quad = 0 \quad (6)$$

Rearranging,

$$V_{\infty} \left( \alpha - \frac{dz}{dx} \right) = \int_0^c \frac{\gamma(\xi) d\xi}{2\pi(x - \xi)} \quad (7)$$

Equation 7 is the FUNDAMENTAL EQUATION FOR THIN AIRFOIL THEORY.

### APPLICATION OF THIN AIRFOIL THEORY TO SYMMETRIC AIRFOIL

Lets apply Equation 7 to a symmetric airfoil. The symmetric airfoil has no camber. Therefore, Equation 7 becomes,

$$V_{\infty} \alpha = \quad (8)$$

For easy simplification of this equation, we need to convert Equation 8 to polar coordinates. We can do that by considering Figure 5.

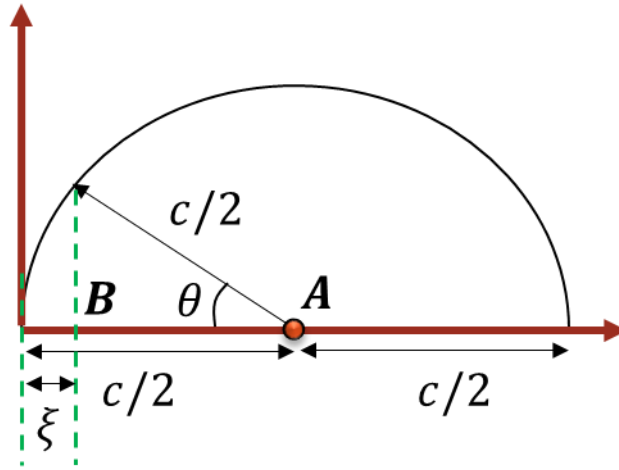


Figure 18 Converting to Polar Coordinates

From Figure 5,

$$AB = \quad (9)$$

$$\xi = \quad (10)$$

$$d\xi = \quad (11)$$

Let

$$x = \quad (12)$$

Substituting Equation 9 – 12 in Equation 8,

$$V_{\infty} \alpha = \quad (13)$$

The analytical solution to Equation 14 is,

$$\gamma(\theta) = \quad (14)$$

The net circulation from vorticity sheet can be represented by,

$$\Gamma = \int_0^c \gamma(\xi) d\xi \quad (15)$$

In polar coordinates, (using Equation 11)

$$\Gamma = \quad (16)$$

Substituting Equation 14 in Equation 16,

$$\Gamma = \quad (17)$$

Simplifying Equation 17,

$$\Gamma = \quad (18)$$

Lift from KJ theorem is,

$$L' = \quad (19)$$

Substituting Equation 18 in Equation 19,

$$L' = \quad (20)$$

The sectional coefficient of lift is,

$$C_l = \quad (21)$$

Substituting Equation 20 in Equation 21 and simplifying,

$$C_l = \quad (22)$$

The lift curve slope is,

$$C_l = \quad (23)$$

## QUIZ 1

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Dear students,

Please don't freak out when you hear quiz! This quiz doesn't play any role in your grade or class performance. I just wanted to know how much information you were able to retain without memorizing or preparing for the test. It is a way to assess my teaching style and class structure. This quiz contains just 10 multiple choice questions. I hope you will be able to answer them all in 10 minutes.

---

Name:

In each question, please state the level of confidence (0 – 100%) in your answers.

1. Kelvin's theorem states

- a) Circulation is zero
- b) Circulation changes with time
- c) Rate of change of circulation with time is zero
- d) None of the above

2. Helmholtz 2<sup>nd</sup> theorem states

- a) Circulation is constant
- b) Vortex tubes doesn't end in a fluid
- c) Vortex tubes can end in a fluid
- d) Circulation is zero

3. Negative pressure coefficient means

- a) Velocity is increasing
- b) Velocity is decreasing
- c) Velocity is zero
- d) Velocity is equal to freestream velocity

4. Thrust equation can be obtained from

- a) Conservation of mass
- b) Conservation of mass and Conservation of momentum
- c) Conservation of momentum
- d) Conservation of energy

5. Smoke lines in a flow visualization experiment in the wind tunnel represents

- a) Streamlines
- b) Streaklines
- c) Pathlines
- d) None of the above

6. A flow can be irrotational even though velocity gradients are present

- a) True
  - b) False
7. Bernoulli's equation can be used in a rotational flow
- a) True
  - b) False
8. Crocco's theorem states
- a) Viscosity produces vorticity in the flowfield
  - b) Vorticity changes the total pressure in the flowfield
  - c) Vorticity changes the static pressure in the flowfield
  - d) Vorticity changes the dynamic pressure in the flowfield
9. Vorticity can be produced by
- a) Viscous forces
  - b) Body forces
  - c) Misalignment between pressure and density gradients
  - d) Velocity gradients
  - e) All of the above
10. Winglets reduce the circulation of the wingtip vortex significantly
- a) True
  - b) False

## QUIZ 2

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Dear students,

Same as before, please don't freak out when you hear quiz! This quiz doesn't play any role in your grade or class performance. I just wanted to know how much information you were able to retain without memorizing or preparing for the test. It is a way to assess my teaching style and class structure. This quiz contains just 10 multiple choice questions. I hope you will be able to answer them all in 10 minutes.

---

Name:

In each question, please state the level of confidence (0 – 100%) in your answers.

1. We were able to add different elementary flows to simulate complex flows because

- e) Elementary flows satisfy Laplace's Equation
- f) Elementary flows doesn't satisfy Laplace's Equation
- g) Elementary flows satisfy conservation of momentum
- h) None of the above

2. The stream function equation satisfies

- e) Conservation of mass
- f) Conservation of momentum
- g) Conservation of energy
- h) None of the above

3. Kutta condition states that

- e) Vorticity at the trailing edge should be zero
- f) Circulation at the trailing edge should be zero
- g) Streamlines can bend around the trailing edge
- h) Streamlines can intersect at the trailing edge

4. Thin airfoil theory determines

- e) Circulation distribution on the camber line
- f) Vorticity distribution on the camber line
- g) Vorticity distribution on the chord line
- h) Circulation distribution on the chord line

5. Smoke lines in a flow visualization experiment in the wind tunnel represents

- e) Streamlines
- f) Streaklines
- g) Pathlines
- h) None of the above

6. Center of pressure moves with angle of attack for a cambered airfoil
- c) True
  - d) False
7. Center of pressure moves with angle of attack for a symmetric airfoil
- c) True
  - d) False
8. One of the boundary conditions for the thin airfoil theory is
- e) The tangential velocity component on the surface should be zero
  - f) The normal velocity component on the surface should be zero
  - g) Vorticity distribution is zero
  - h) Vorticity distribution is constant
9. Lift curve slope of a symmetric and cambered airfoil is (select all that apply)
- f)  $2\pi$  rad inverse
  - g)  $2\pi$  degree inverse
  - h) 0.11 rad inverse
  - i) 0.11 degree inverse
10. Panel methods determine (select all that apply)
- c) Coefficient of pressure on the surface of the object
  - d) Pressure distribution in the flow around the object
  - e) Tangential velocity on the surface of the object
  - f) Vorticity on the surface of the object